



ESKISEHIR OSMANGAZI UNIVERSITY
ELECTRICAL AND ELECTRONICS ENGINEERING
2021-2022 SPRING SEMESTER
DIGITAL SYSTEMS LABORATORY

EXPERIMENT 3
ADDER / SUBTRACTOR



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GROUP II – D



EXPERIMENT 3 REPORT

OBJECTIVES

- The addition and subtraction of 4-bit binary adder numbers is studied using the integrated circuit.

INTRODUCTION

4 bit binary adders subtraction, addition and binary numbers signed and unsigned representation on the circuit.

PROCEDURE

Q1- Fill the table with the 1's and 2's complements of the given binary numbers.

	1s Complement	2s Complement
1011	0100	0101
1111	0000	0001
0101	1010	1011
0000	1111	0000
0011	1100	1101

Q2- Fill the table with the 8-bit signed representations of the decimal numbers.

Decimal	Signed Magnitude	Signed 1s complement	Signed 2s complement
-55	11001001	00110110	00110111
-23	11101001	00010110	00010111
-15	11110001	00001110	00001111
-4	11111100	00000011	00000100
-105	10010111	01101000	01101001

Q3- The 4 bit adder/subtractor circuit implemented with IC 7483 is shown in Figure 1. Explain the operation of the circuit with your own words, briefly.

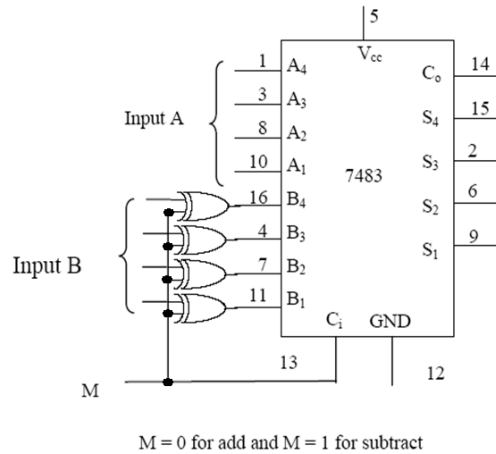


Figure 1

Q4- What is the difference between “*half adder*”, “*full adder*”, “*ripple carry adder*” and “*carry look ahead adder*” circuits? Explain with your own words, briefly.

The half adder produces the sum of the two inputs.

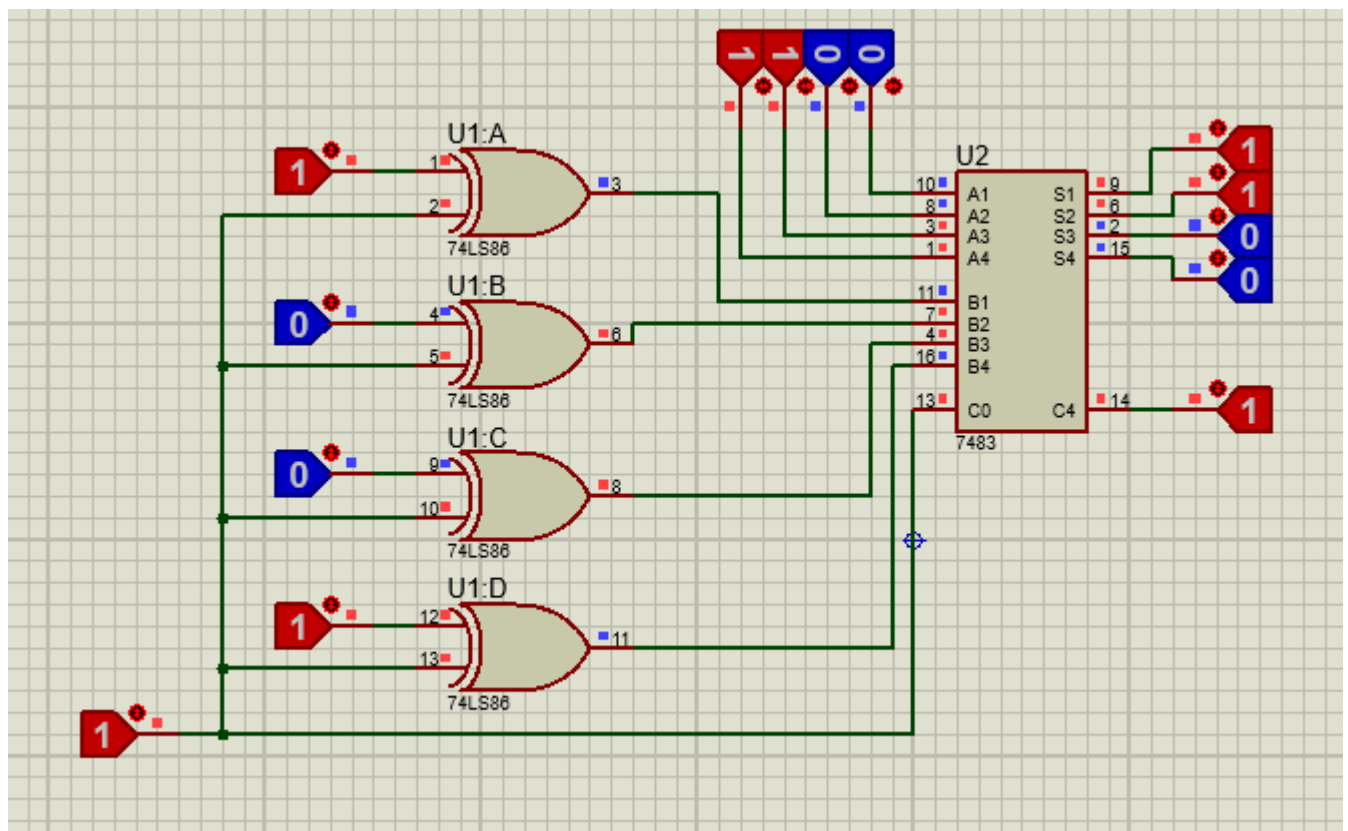
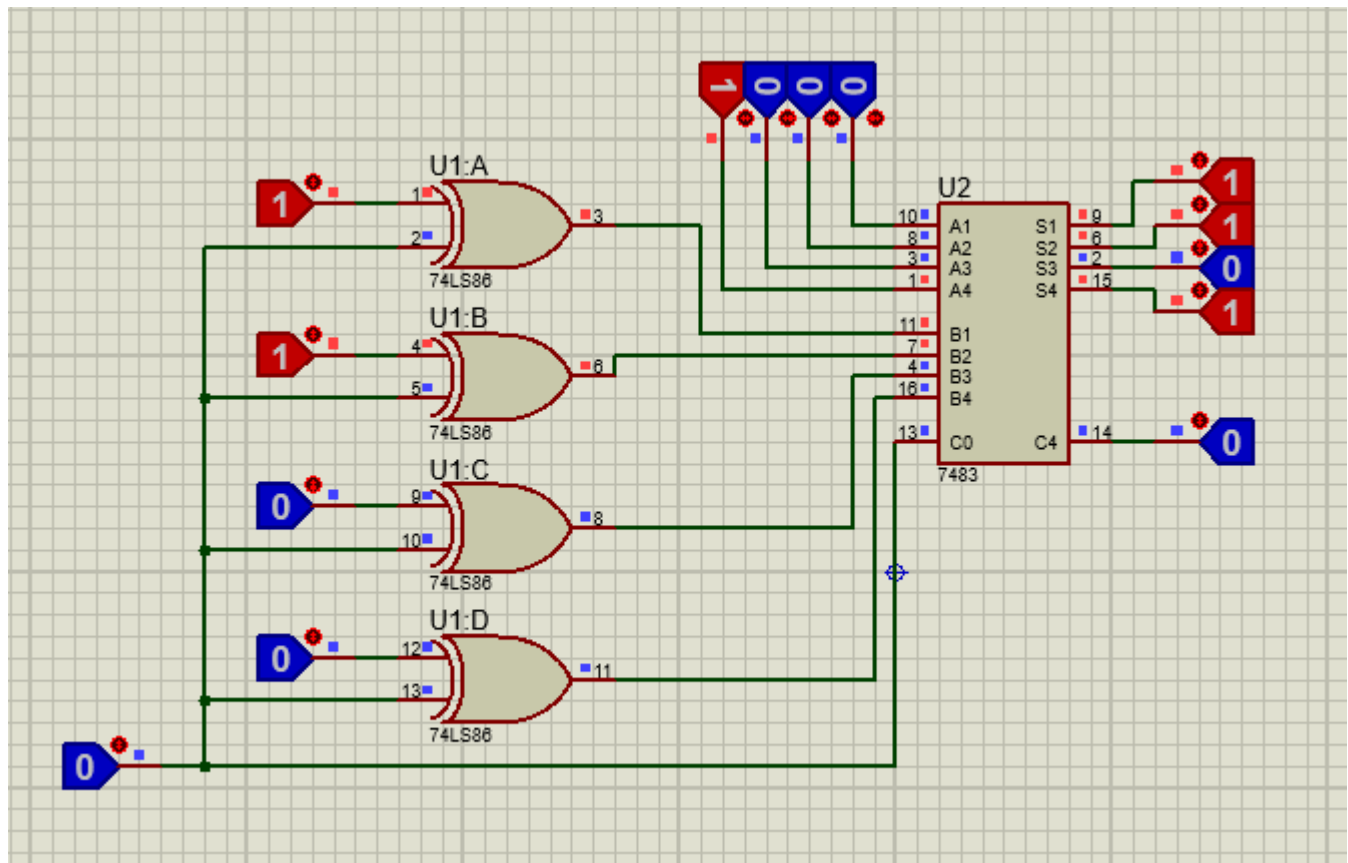
The full adder produces the sum of the three inputs and the carry value.

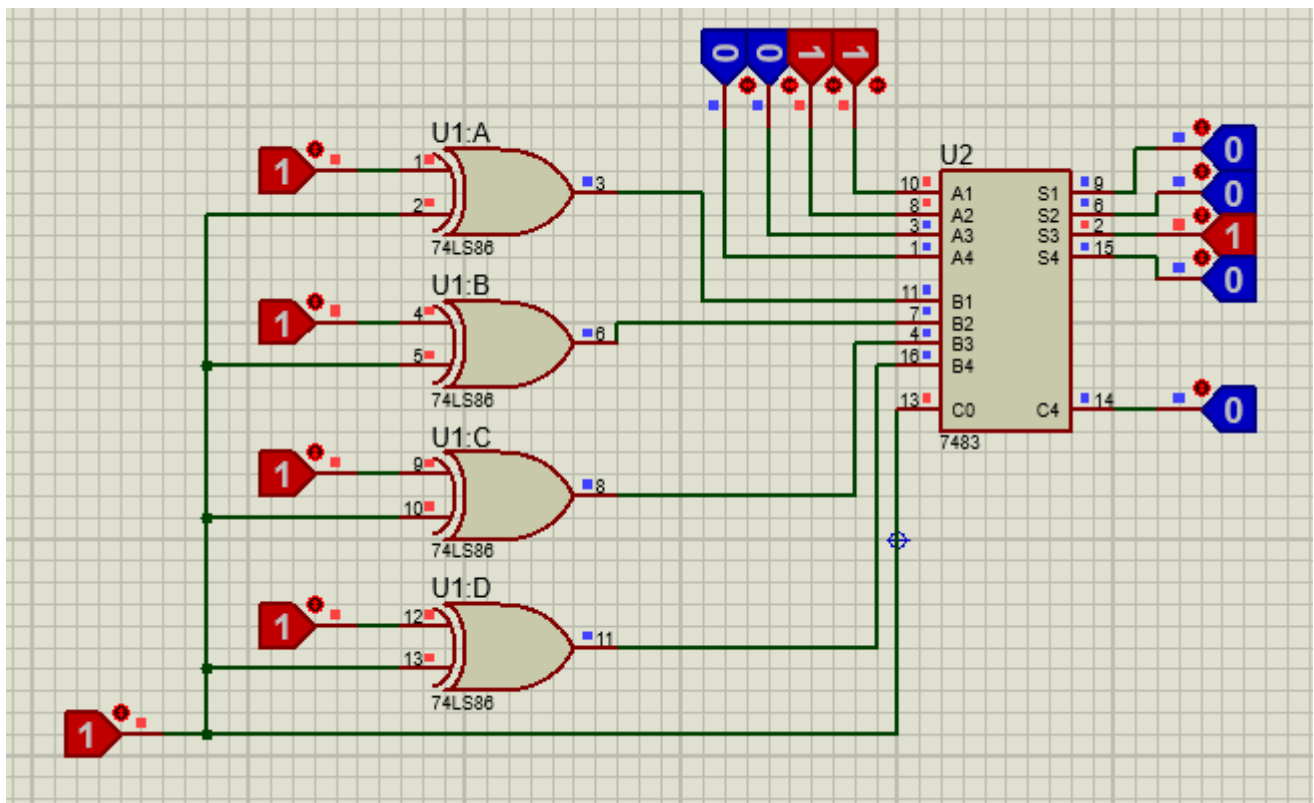
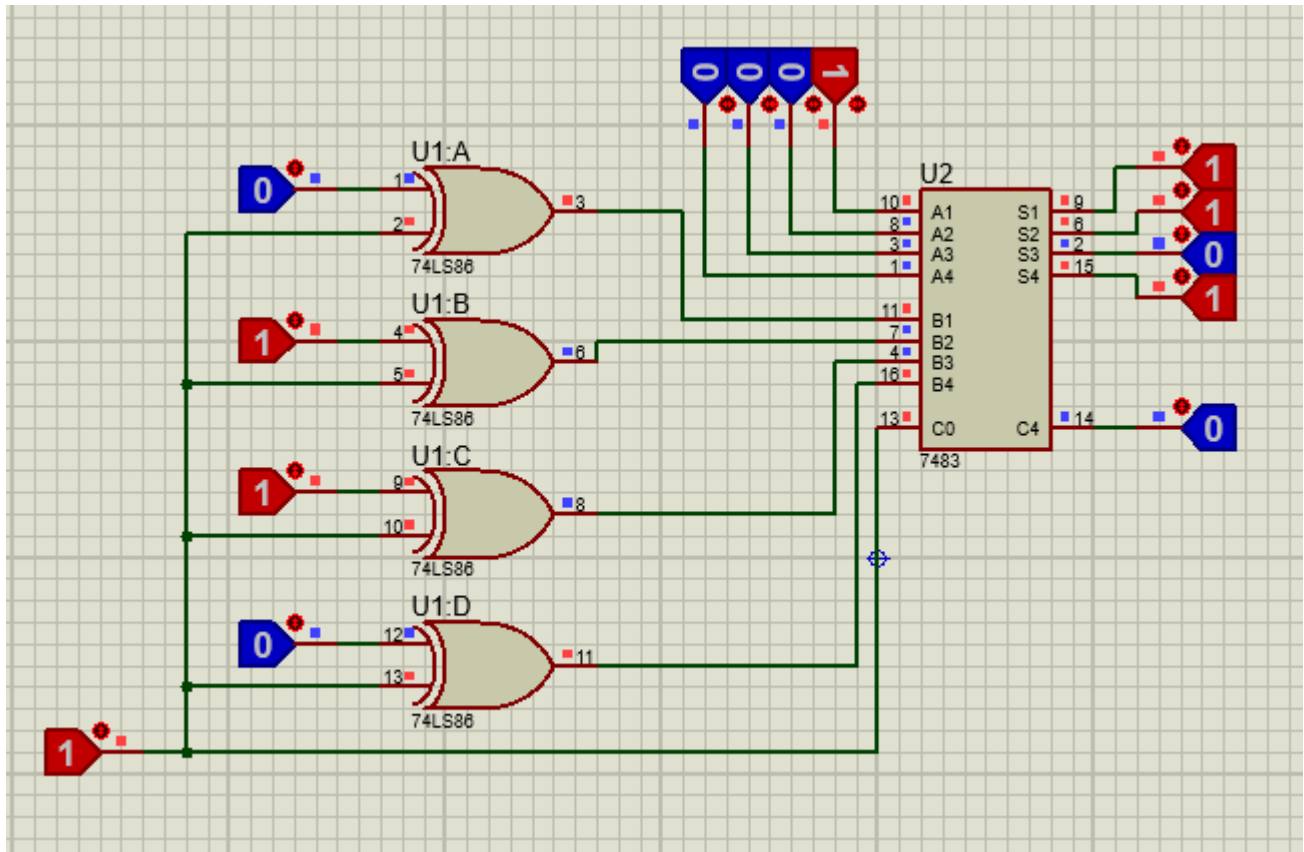
The ripple carry adder contain individual single bit full adders which consist of 3 inputs and 2 outputs.

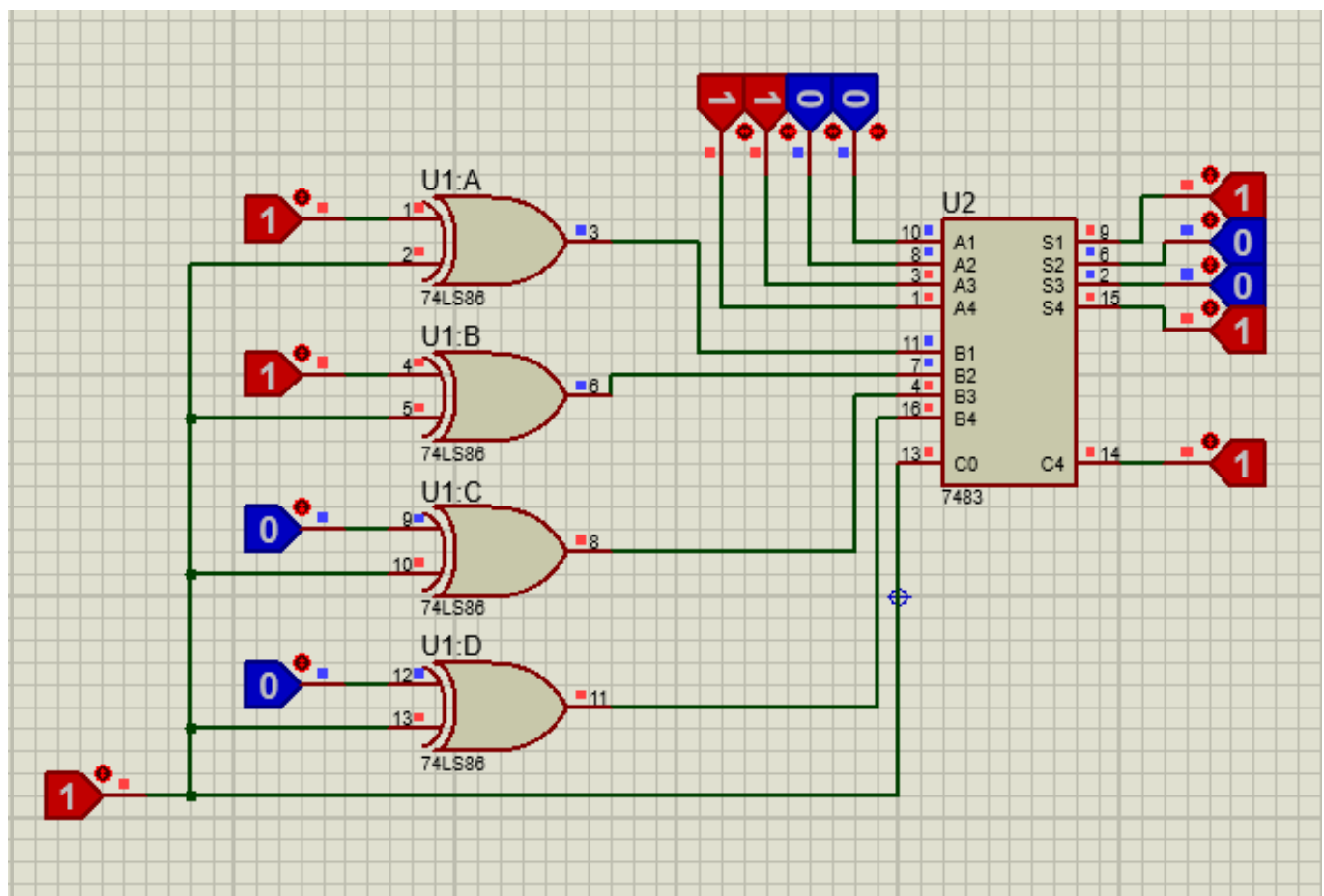
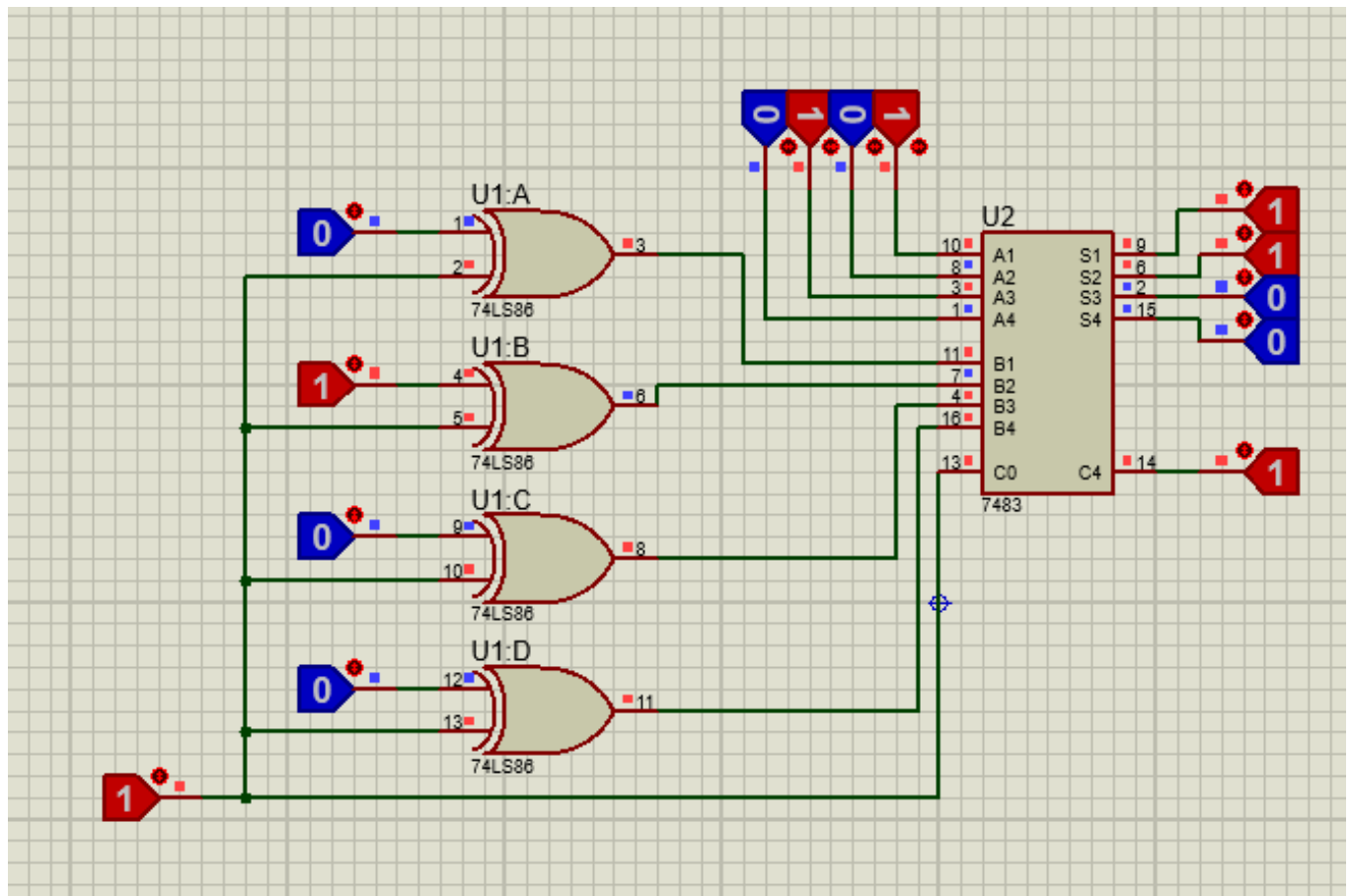
These full adders are connected together in cascade form to create a ripple carry adder.

A carry-look ahead adder (CLA) is a type of electronics adder used in digital logic. A carry-look ahead adder improves speed by reducing the amount of time required to determine carry bits.

Q5- Connect the adder / subtractor circuit given in Fig.1 in your simulation program. Apply the following numbers in 4-bit binary to circuit and select the operation with the select (M), and fill the below table with obtained sum (S) and carry out (C0) values. You must add your circuit drawing and circuit outputs for each input to your report.







Decimal	Sum				Carry Out
A + B	S3	S2	S1	S0	C0
8 + 3	1	0	1	1	0
12 - 9	0	0	1	1	1
-2 + 5	0	0	1	1	1
3 - 15	1	1	0	0	0
(-6) - (-1)	0	1	0	1	0
(-3) - (-12)	1	0	0	1	1

Q6- Comment on “Select” (M) and “Carry Out” (C0) for each operation performed in Q5.

1st:Mod=0 carry out=0

2nd:Mod=1 carry out=1

3rd:Mod=1 carry out=1

4th:Mod=1 carry out=0

5th:Mod=1 carry out=0

6th:Mod=1 carry out=1

DISCUSSION AND CONCLUSIONS

Addition and subtraction processes were implemented for various values with adder and subtractor circuits. I could not get results because I could not fully integrate the cables into the circuit in the classroom. For this, I added the results of the circuit I made over proteus.